



Validation Workflow Description: Absolute Radiometric Calibration

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1 Introduction

1.1 Scope

This document describes a Validation Workflow for the Absolute Radiometric Calibration uncertainty specified by L1 Optical EO products, developed to support the EODH Data Product Quality Assurance Process [AD-1]. The purpose is to define the validation method and demonstrate it has been developed following processes in line with best practice, as defined by the ESA/NASA EO Product QA Framework [RD-1]. This framework defines requirements for a variety of aspects for validation methods – awarding an overall grade of *Basic*, *Good*, *Excellent*, or *Ideal*.

The document is structured as follows:

- Section 2 describes a summary of the Validation Workflow and its objectives.
- Section 3 reviews the validation method for each of the aspects defined by the ESA/NASA EO Product QA Framework.
- Section 4 describes the validation methodology in detail.
- Section 5 describes the implementation architecture in the EODH system.

This Validation Workflow has initially been prepared as a demonstrator to pilot the Data Product Validation component of the Data Product QA Process within the EODH Pathfinder project. In this pilot phase, this workflow will be run for three optical datasets with the EODH Data Product Catalogue: Sentinel-2 L1C, Planet SuperDove L3B, and Airbus Pléiades MS-ORTHO.

1.2 Changes Log

Issue	Date	Description
0.1	24/03/25	Initial draft for review
1.0	23/04/25	First version for pathfinder

1.3 Applicable Documents

[AD-1] EODH QA Service, 'Data Product Quality Assurance Process', 2025.

1.4 Reference Documents

- [RD-1] Hunt et al., 'Earth Observation Mission Quality Assessment Framework' v2.2, EDAP Project Document, 2021. [Available Online](#).
- [RD-2] Bouvet, M.; Thome, K.; Berthelot, B.; Bialek, A.; Czapla-Myers, J.; Fox, N.P.; Goryl, P.; Henry, P.; Ma, L.; Marcq, S.; Meygret, A.; Wenny, B.N.; Woolliams, E.R. RadCalNet: A Radiometric Calibration Network for Earth Observing Imagers Operating in the Visible to Shortwave Infrared Spectral Range. Remote Sens. 2019, 11, 2401, <https://doi.org/10.3390/rs11202401>
- [RD-3] RadCalNet Guidance Uncertainty Analysis, June 2018. https://www.radcalnet.org/documentation/RadCalNetGenDoc/G4-RadCalNetGuidance-UncertaintyAnalysis_V1.pdf
- [RD-4]

OGC Best Practice for Earth Observation Application Package. 2021.
Accessed at: <https://docs.ogc.org/bp/20-089r1.html>

1.5 Acronyms & Abbreviations

Cal/Val	Calibration/Validation
CEOS	Committee for Earth Observation Satellites
EDAP	Earthnet Data Assessment Project
EO	Earth Observation
EODH	Earth Observation Data Hub
ESA	European Space Agency
NASA	National Aeronautics and Space Administration
NCEO	National Centre for Earth Observation
NPL	National Physical Laboratory
QA	Quality Assurance
RadCalNet	Radiometric Calibration Network
ROI	Region of Interest
TOA	Top-of-Atmosphere

2 Absolute Radiometric Calibration Uncertainty

Absolute radiometric calibration is the processing of optical Earth Observation (EO) imagery from the raw data measured by the instrument to physical units, such as radiance or reflectance. It is the fundamental step in ensuring the data products are quantitatively meaningful and consistent across different sensors, locations, and time periods.

A sensor's calibration is established through comparison with well-known reference sources – typically a combination of pre-flight laboratory measurements, onboard calibration devices (if available), and “vicarious” targets such as ground-based instrumentation. The absolute radiometric calibration uncertainty defines the performance of the calibration process and is a significant driver of data's fitness-for-purpose for many quantitative applications. Independent validation of a product's achieved absolute radiometric calibration uncertainty is therefore essential to build user confidence that the data is suitable for their needs.

Radiometric validation entails comparing satellite data to a set of independent calibration reference measurements. The EO cal/val (calibration/validation) community have developed a suite of high-quality reference measurements to support this activity in recent years – notably this includes the RadCalNet (Radiometric Calibration Network) initiative [RD-2]. RadCalNet is a global network of automated, ground-based calibration sites that provide continuous, standardised measurements of top-of-atmosphere (TOA) for validating. Managed by CEOS (Committee for Earth Observation Satellites), RadCalNet offers free, high-quality, uncertainty-quantified reference data from sites with stable surface properties, such as deserts and dry lakebeds (see Figure 1).

The Validation Workflow defined in this document aims provide the automated, real-time validation of optical EO data products against RadCalNet – providing a simple performance verification to users through “QA Check Results” (as pass/partial/fail) for EODH Data Product Catalogue collections.

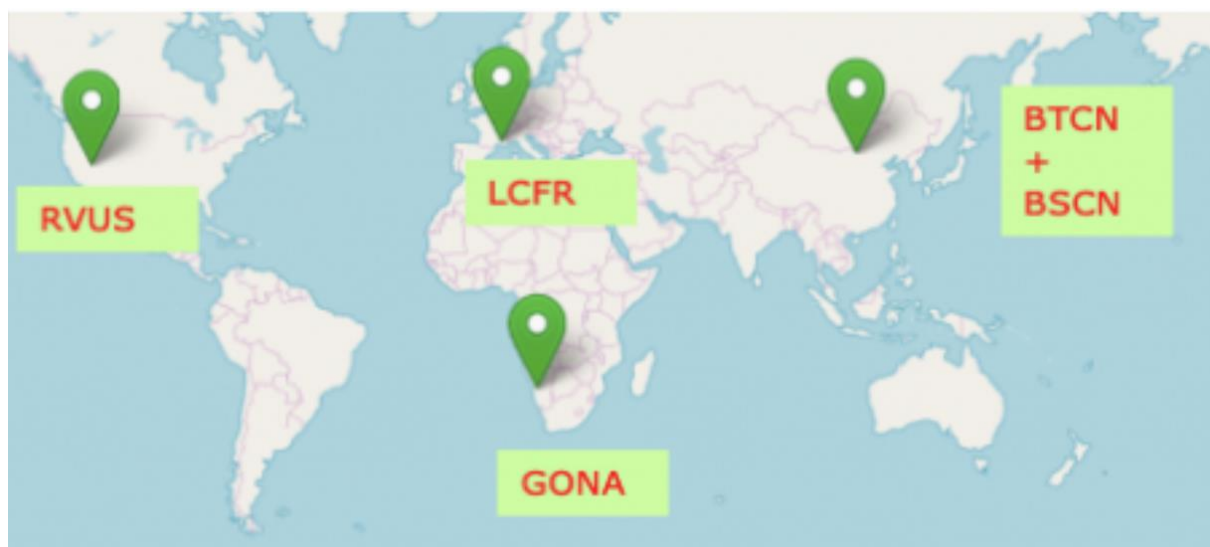


Figure 1 - Locations of RadCalNet sites

3 Validation Workflow Quality Analysis

This section demonstrates the way in which this Validation Workflow it has been developed in line with best practice, as defined by the ESA/NASA EO Product QA Framework [RD-1]. This framework defines requirements for a variety of aspects for validation methods – awarding an overall grade of *Basic*, *Good*, *Excellent*, or *Ideal*. These aspects are:

- Validation Methodology – the fitness-for-purpose of the method applied to estimate product for performance.
- Reference Data Quality – (*where applicable*) the extent to which any reference data used for comparison in the validation is of a commensurate quality.
- Validation Representivity – the extent to which the validation results are applicable to the full range of observations the satellite may make (e.g., dynamic range, geographic extent, surface biomes)

The tables below cover each of these topics in turn, providing justification for the grade achieved with reference to the underpinning evidence for this (presented either in this document or reference documents).

Validation Methodology	
Grade: Excellent	
Justification	<i>The validation methodology, described in Section 4, shows the various processing steps in detail to enable the comparison of the satellite mission data with the in-situ reference.</i> <i>The method takes into account the uncertainties of the satellite itself, the reference product and the comparison process, however the uncertainty due to the comparison process could be refined.</i>
References	• See Section 4

Reference Data Quality	
Grade: Excellent	
Justification	<i>Becoming an official RadCalNet site is a rigorous process, and the site must meet specific requirements. These requirements include that the data is SI-traceable, has documented uncertainties and that it adheres to the Quality Assurance Framework for Earth Observation (QA4EO). This includes fully quantifying the uncertainty associated with the input parameters and assumptions used in all the processing involved. See [RD-3] for the full outline on the uncertainty analysis of the data to be undertaken.</i> <i>All RadCalNet sites undertake the same processing from raw data to TOA reflectances.</i>

	<i>For commercial missions with uncertainties between 5-10 % RadCalNet is a suitable reference. However, any missions claiming <5% cannot be validated to this level with RadCalNet as its own uncertainties are higher than this.</i>
References	• RD-2 • RD-3

Validation Representivity	
Grade: Good	
Justification	<i>For optical data validation, comparisons with RadCalNet sites, specifically the GONA desert site in Gobabeb, Namibia, are widely used within the community.</i> <i>However, this is not fully representable of the entire satellite mission dataset, as it is currently confined to a comparison with one RadCalNet site (e.g., dynamic range). In future, to achieve ideal here, the development of more RadCalNet sites of various surface types would increase the representivity of this validation of the mission data as a whole.</i>
References	• RD-2

4 Validation Methodology Description

Radiometric validation is achieved through a comparison of the satellite mission L1 product with RadCalNet in-situ data. RadCalNet data is collected at well-characterised and homogeneous sites and is nadir-view TOA reflectance at 30-minute intervals between 9am-3pm local standard time, with a spectral range of 400-2500 nm, at 10 nm intervals. The satellite mission data products considered are multispectral data. For the pathfinder phase of EODH the satellite sensors under test have the spectral range and spatial resolutions outlined in Table 1.

Satellite Mission	Spectral range	Spatial resolutions
Sentinel-2	~440-2200 nm	10 m, 20 m, 60 m, depending on the band
Planet SuperDove	~430-885 nm	3.7 m
Airbus Pléiades	~450-915 nm	2 m

Table 1: Spectral range and spatial resolutions for the three satellite sensors under test during the pathfinder phase of the EODH project.

The validation is carried out via a processing chain made up of the following steps:

- *Step 1: Identify Matchups* – the process by which coincident data from the satellite sensor under test and the in-situ reference are identified.
- *Step 2: Prepare Data for Comparison* – process both the data from the satellite sensor and the in-situ reference data, through temporal interpolation, spectral band adjustment and spatial aggregation
- *Step 3: Evaluate QA Check Result* – perform comparison between processed satellite and reference measurements and interpret as a QA Check Result (i.e., pass/partial/fail).

The following defines the processing for each of these steps in turn.

4.1 Step 1: Identify Matchups

Figure 2 outlines the process for Step 1, where matchup events are identified between satellite sensors and RadCalNet from overpasses of the satellite over the in-situ site. This is achieved by making use of the satellite catalogue APIs to search for the satellite data over that ROI (region of interest). Once the relevant products are identified and retrieved, e.g. via the EODH API for the satellite products, the regions of collocated observations are extracted. This data is stored in a “collocation dataset”.

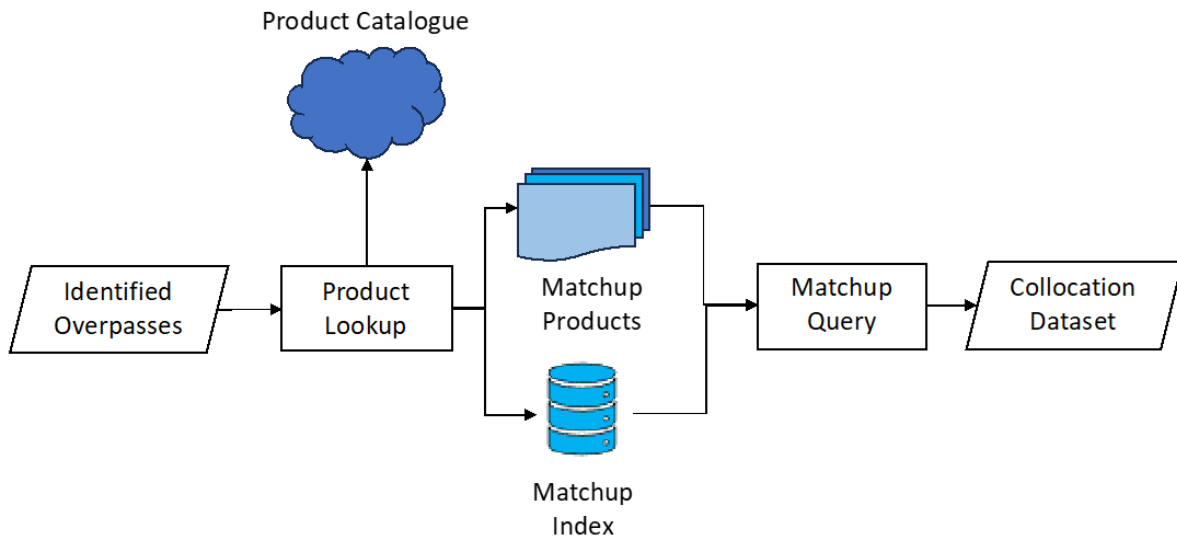


Figure 2: Step 1 - Matchup generation.

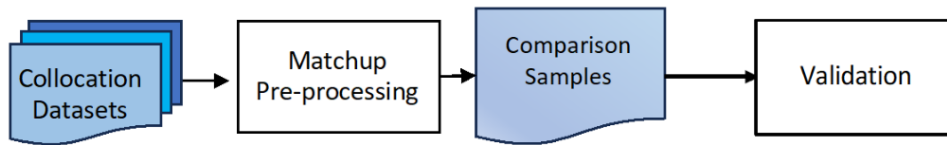


Figure 3: Steps 2 and 3 - Comparison and validation.

4.2 Step 2: Prepare Data for Comparison

The collocation datasets prepared in Step 1 are the inputs to Step 2, shown Figure 3. This step carries out the processing to make these datasets comparable, outputting “comparison samples”. To create these comparison samples, there are several characteristics that need to be considered, such as the varying spectral responses of the two sensors, the difference in viewing geometry and the difference in spatial resolutions. To correct for these, the matchup pre-processing step includes spectral band adjustment and spatial aggregation, as well as compensating for viewing geometry mismatch (changes in surface reflectance, polarization), and temporal mismatch (changes in illumination angle, atmosphere).

These outputs are stored in a “matchup dataset” which is then used for validating the satellite sensor against the reference site. The full description of the Optical Radiometric Validation Method used for this Validation Workflow is outlined in Section 4.3.

4.2.1 Processing for Reference Data

Step 1: Spectral band integration

The first step of the matchup processing is to integrate the high resolution spectrum of the reference within the collocation region, $\rho_{RCN}(\lambda_{RCN})$, so that it matches that of the sensor under test, $\rho_{RCN}(\lambda_s)$. Each target sensor spectral band is defined by its SRF, $SRF_{s,i}(\lambda_{SRF_{s,i}})$.

The method used to achieve this spectral adjustment has the following steps for each band:

a) Interpolate the reference reflectance spectrum

The reference reflectance spectrum must be over sampled to match the wavelength grid of the target sensor SRF, i.e., to evaluate $\rho_{RCN}(\lambda_{SRF_{s,i}})$. This is necessary for the numerical integration in the following step. This is performed using PCHIP interpolation.

The spectral range of this oversampling should be limited to the wavelengths around the non-zero region of the target sensor SRF to limit the computation and avoid numerical errors.

b) Perform numerical spectral band integration

Per band of the target sensor $\rho_{RCN}(\lambda_{SRF_{s,i}})$ evaluate the spectral band integral, performed using the trapezium rule, as follows:

$$\rho_{RCN}(\lambda_{s,i}) = \frac{\int \rho_{RCN}(\lambda_{SRF_{s,i}}) SRF_{s,i}(\lambda_{SRF_{s,i}}) d\lambda}{\int SRF_{s,i}(\lambda_{SRF_{s,i}}) d\lambda}$$

Equation 1: Spectral band integration.

Step 2: Temporal interpolation

The reference reflectance data, collected every half hour, is temporally interpolated to the time of acquisition of the satellite sensor, which calculated per band as follows:

$$\rho_{RCN} = \frac{\rho_{RCN_0}(t_1 - t) + \rho_{RCN_1}(t - t_0)}{t_1 - t_0}$$

Equation 2: Temporal Interpolation.

Where ρ_{RCN_i} is the reference reflectance at the time collected, t_i , and t is the time of acquisition of the satellite sensor.

4.2.2 Processing for Satellite Data

Step 1: Spatial aggregation

Given that the reference in-situ point data is representative of the whole ROI – as the site is selected based on its homogeneity – the satellite mission data is spatially averaged, per pixel, over the ROI.

4.3 Step 3: Evaluate QA Check Results

The validation method takes the output of the comparison routine outlined above and calculates a pass/partial/fail result based on how well the satellite data agrees with the in-situ reference data – considering the uncertainties from each the satellite data product, the RadCalNet in-situ product and the comparison itself.

The comparison routine outputs the reflectance bias, per band, between the satellite and in-situ reference data and the corresponding uncertainties for that comparison. In addition, the RadCalNet reference product contains uncertainties for each reflectance measurement and the satellite data uncertainties are taken as what is quoted in the relevant documentation supplied by the data provider.

Using this information, the Equivalence (E) is calculated via the following per set timeframe (e.g. per month), per band:

$$E = \frac{\rho_a - \rho_b}{\sqrt{u^2(\rho_a) + u^2(\rho_b) + u^2(comp)}}$$

Equation 3: Equivalence ratio.

Where ρ_a is the satellite data reflectance, ρ_b is the in-situ data reflectance, with corresponding uncertainties $u(\rho_a)$ and $u(\rho_b)$, respectively, and $u(comp)$ is the uncertainty arising from the comparison process.

The result E is calculated per band for the average bias in each set timeframe (e.g. per month), where:

$E \leq 1$ defines a ‘pass’

$E \leq 2$ defines a ‘partial pass’

$E > 2$ defines a ‘fail’

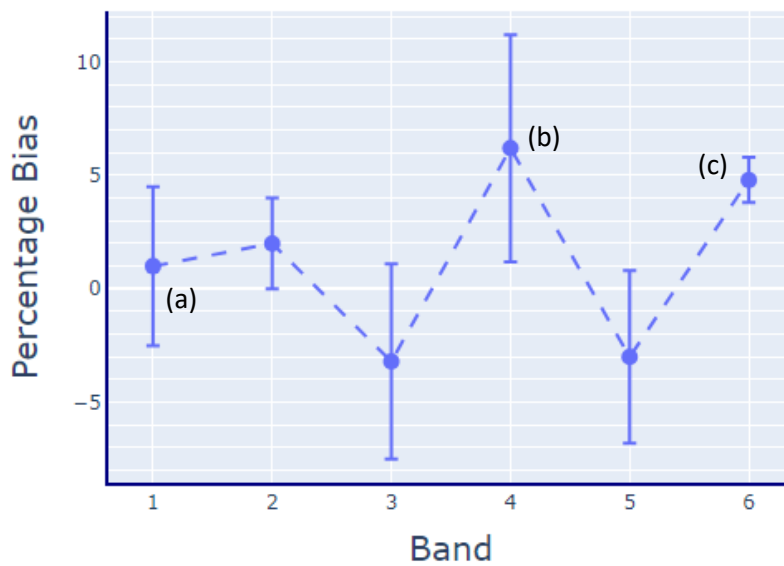


Figure 4: Visual representation of the check results.

This can be seen visually in Figure 4, where $E \leq 1$ refers to the uncertainty bars crossing zero (a), $E \leq 2$ represents the uncertainty bars *not* crossing zero but zero is within twice the uncertainty (b), and $E > 2$ is where the uncertainty bars do not cross zero and zero is out of this range (c). Therefore, for the example in Figure 4, all bands would pass except for (b) which would achieve partial pass and (c) which would fail.

The overall result given for the whole collection is based on how many bands pass: i.e., if all bands pass, the data collection receives a 'pass' grade, if they all fail, the collection receives a 'fail' overall, if there is a combination of bands passing, failing and partially passing, the overall grade awarded is 'partial pass'.

The results, including the results per band, are displayed on the EODH under the Quality Assurance tab for that data collection.

5 Implementation Architecture

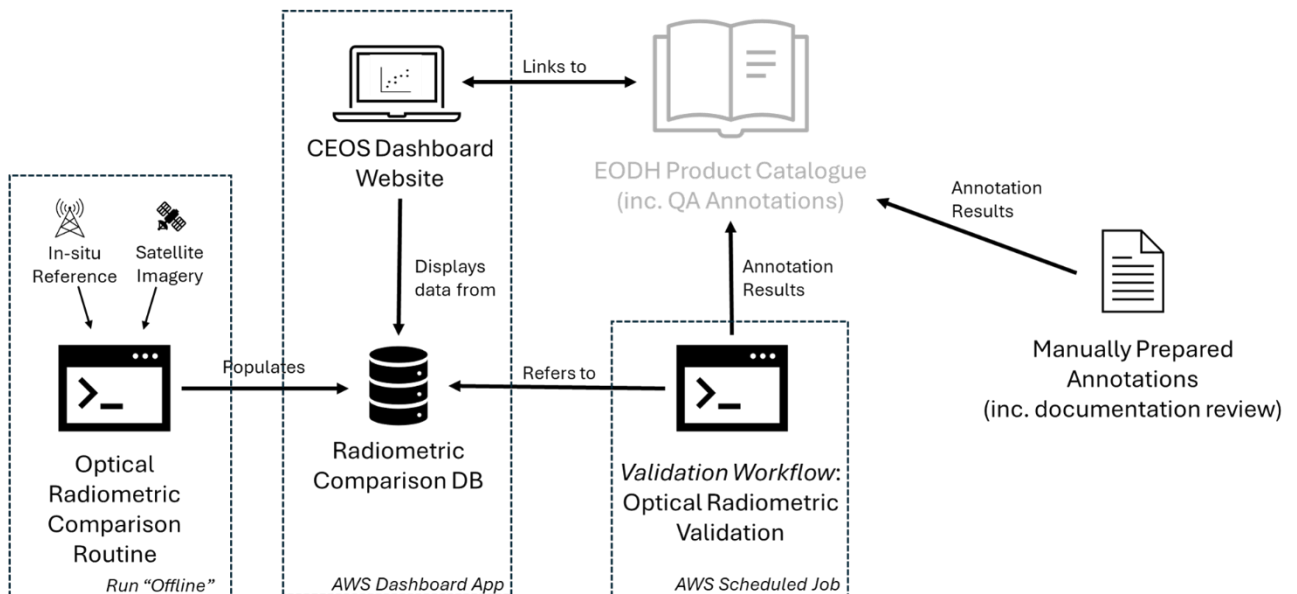


Figure 5: Absolute Radiometric Calibration uncertainty validation software architecture

This section defines the software architecture implemented to run the absolute radiometric calibration uncertainty validation workflow, as developed within the EODH Pathfinder phase. This architecture is made up of four main components:

- i) **Optical Radiometric Comparison Routine** that is run “offline” – carried out on 3 optical data collections available on the Hub, comparing to an in-situ reference site.
- ii) **AWS Dashboard App** comprising of the Radiometric Comparison Database (populated with outputs from the Comparison Routine and relevant metadata for the data collections) and the CEOS Dashboard website.
- iii) **Validation Workflow** running periodically via an AWS scheduled job, reading from the Radiometric Comparison Database and outputting QA Optical Radiometric Validation results for ingestion as annotations into the EODH Product Catalogue.
- iv) **Manually prepared annotations** from offline assessment that are provided directly to TPZ for ingestion into the EODH Product Catalogue (e.g. documentation reviews, EOIS dataset quality results).

5.1 Optical Radiometric Comparison Routine

This routine is run offline and compares each of the satellite data collections against an in-situ reference site, specifically, the RadCalNet desert site in Gobabeb (GONA). This site in particular was chosen for its radiometric stability and regular data acquisition. We define a matchup as when a satellite passes over the site, and each matchup is found and processed following the full method for comparison which is described below, in Section 3.1.

The outputs of the routine are added to the Radiometric Comparison Database, deployed on AWS, from which the data is extracted for display in the CEOS Calibration Dashboard website and to be used in the calculation of the validation workflow results.

5.2 CEOS Calibration Dashboard Website and Radiometric Comparison Database

The first beta version of the CEOS Calibration Dashboard Web App is deployed on AWS and will allow users to explore in more detail the data behind the QA Optical Radiometric Validation checks. As mentioned previously, it is populated by the Radiometric Comparison Database, also deployed on AWS via a secure connection.

The interactive Dashboard displays a timeseries of matchups between a satellite mission and various RadCalNet in-situ sites, based on the selection made by the user, including bands and timeframe of interest. For a more in-depth look, the user can select a specific point of interest from the plot, referring to one matchup between the selected satellite mission and reference site. This will populate the “Matchup Analysis” panel with more detailed information for that matchup, including solar azimuth, satellite viewing angle, cloud percentage and AOD. In addition, there is a scene location image and a secondary plot of the satellite vs in-situ reflectances, per band.

The Dashboard will be linked to directly from the EODH catalogue entry for a given data collection, with a pre-defined selection input into the timeseries display, based on a dynamic URL created using the data collection name. Moreover, if applicable, there will be a link from the Dashboard back to the EODH catalogue data collection page, based on the satellite mission selected.

5.3 Validation Workflow: Optical Radiometric Validation

The validation workflow is written using CWL (Common Workflow Language) and follows OGC Best Practices for EO Workflows (see [RD-4]). The workflow is deployed on AWS and scheduled to run routinely based on a *cron*. job script monthly, where it calls the Radiometric Comparison Database to obtain the relevant comparison outputs which are then used to evaluate the pass/partial/fail status per collection.

The output of each validation workflow check will consist of a pass/partial/fail result (See Section 3.2 for the method in which the result is calculated from the optical radiometric comparison routine output) and other information to be logged, e.g. name of collection checked, name of check, date of check, date range of validity for that check. This output will be in the form of a JSON blob – see [RD-2] for more detail on the whole check output format – which will be ingested by TPZ, the Hub Developers, as annotations to the EODH catalogue.